

Abstract :

The Data Download Duplication Alert System is designed to prevent unnecessary duplicate downloads of files and datasets within an organization or platform.

Introduction:

In many organizations, users frequently download the same files multiple times, resulting in wasted network bandwidth, increased server load, and duplicate data storage.

Objectives:

1. Detect duplicate file download requests.
2. Maintain a secure download history database.
3. Generate alerts for repeated downloads.
4. Reduce unnecessary bandwidth usage.

Problem Statement:

Organizations often experience repeated downloads of identical files due to lack of visibility into previous download activities. This leads to:

1. Increased bandwidth consumption.
2. Higher server workload.

System Architecture:

Modules

User Authentication Module

User registration and login.

Access control and authorization.

File Management Module

Upload and manage files.

Store file metadata.

Download Monitoring Module

Track all download requests.

Record user activity.

Technologies Used:

Frontend

HTML

CSS

JavaScript

Backend

Python (Flask/Django) or Java

Database

MySQL

TOOLS:

Visual Studio Code

Advantages

Reduces duplicate downloads.

Saves network bandwidth.

Improves server performance.

Enhances user awareness.

Provides download analytics.

Easy to integrate with existing systems.

Future Enhancements

Machine learning-based download prediction.

Email and mobile alerts.

Cloud storage integration.

Real-time dashboard monitoring.

Role-based administrative controls.